No.	Status	Date
LNDE-RPT-96-014-DS-3-CW-101	Issued	1996-03-08

UT Thickness Data Sheet - 3-CW-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 3-CW-101, part of the ATU-100. The scope of work involves the collection of baseline thickness data to establish the initial condition of the asset prior to service.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

Technical Specifications

Line 3-CW-101 is an Amine cooler CW constructed from A106 Gr.B. The line has an outer diameter of 88.9 mm and a nominal thickness of 5.49 mm. The design limits for this line are 517 kPa(g) and 49 °C.

The minimum allowable thickness (t-min) for this line is 2.79 mm. Any reading recorded below this threshold is subject to the issuance of an NCR per site rules.

Inspection Methodology

Thickness measurements were performed in accordance with QSCP-NDT-001 Rev 0, the site NDE procedure issued for ATU-100 pre-startup and in-service NDE.

The ultrasonic testing was conducted following the guidelines set forth in ASTM E797 to ensure the accuracy and repeatability of the thickness readings.

Data Integration

The measurements recorded in this data sheet are integrated into LNDE-RPT-96-014, the baseline ultrasonic thickness survey for ATU-100.

This survey covers 118 CMLs for the unit, which is not yet in service as of the date of this report.

Thickness readings, measured 1996-03-08

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	—	—	7.12	—	—	—	2001-03-08	
6-SG-101-02	7.11	3.56	—	—	7.05	—	—	—	2001-03-08	
6-SG-101-03	7.11	3.56	—	—	7.08	—	—	—	2001-03-08	
6-SG-101-04	7.11	3.56	—	—	7.09	—	—	—	2001-03-08	
6-SG-101-05	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-101-06	7.11	3.56	—	—	7.07	—	—	—	2001-03-08	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	—	—	7.14	—	—	—	2001-03-08	
6-SG-102-02	7.11	3.56	—	—	7.11	—	—	—	2001-03-08	
6-SG-102-03	7.11	3.56	—	—	7.04	—	—	—	2001-03-08	
6-SG-102-04	7.11	3.56	—	—	6.99	—	—	—	2001-03-08	
6-SG-102-05	7.11	3.56	—	—	7.13	—	—	—	2001-03-08	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-02	8.18	4.06	—	—	8.20	—	—	—	2006-03-08	
8-TG-101-03	8.18	4.06	—	—	8.23	—	—	—	2006-03-08	
8-TG-101-04	8.18	4.06	—	—	8.10	—	—	—	2006-03-08	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-05	8.18	4.06	—	—	8.22	—	—	—	2006-03-08	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-101-02	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-101-04	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-101-05	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-101-06	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-101-07	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	—	—	6.02	—	—	—	2006-03-08	
4-LA-102-02	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-03	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-102-04	6.02	3.18	—	—	5.99	—	—	—	2006-03-08	
4-LA-102-05	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-06	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	
4-LA-102-07	6.02	3.18	—	—	6.00	—	—	—	2006-03-08	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	—	—	7.02	—	—	—	2006-03-08	
6-LA-103-02	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-03	7.11	3.56	—	—	7.08	—	—	—	2006-03-08	
6-LA-103-04	7.11	3.56	—	—	7.17	—	—	—	2006-03-08	
6-LA-103-05	7.11	3.56	—	—	7.19	—	—	—	2006-03-08	
6-LA-103-06	7.11	3.56	—	—	7.16	—	—	—	2006-03-08	
6-LA-103-07	7.11	3.56	—	—	7.18	—	—	—	2006-03-08	
6-LA-103-08	7.11	3.56	—	—	7.04	—	—	—	2006-03-08	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	—	—	8.52	—	—	—	2006-03-08	
4-RA-101-02	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-101-03	8.56	3.81	—	—	8.61	—	—	—	2006-03-08	
4-RA-101-04	8.56	3.81	—	—	8.63	—	—	—	2006-03-08	
4-RA-101-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-07	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-101-08	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	
4-RA-101-09	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-101-10	8.56	3.81	—	—	8.67	—	—	—	2006-03-08	
4-RA-101-11	8.56	3.81	—	—	8.68	—	—	—	2006-03-08	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-02	8.56	3.81	—	—	8.57	—	—	—	2006-03-08	
4-RA-102-03	8.56	3.81	—	—	8.56	—	—	—	2006-03-08	
4-RA-102-04	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-102-05	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-102-06	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-102-07	8.56	3.81	—	—	8.58	—	—	—	2006-03-08	
4-RA-102-08	8.56	3.81	—	—	8.49	—	—	—	2006-03-08	
4-RA-102-09	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	—	—	11.00	—	—	—	2006-03-08	
6-RA-103-02	10.97	4.57	—	—	11.09	—	—	—	2006-03-08	
6-RA-103-03	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-04	10.97	4.57	—	—	10.98	—	—	—	2006-03-08	
6-RA-103-05	10.97	4.57	—	—	10.95	—	—	—	2006-03-08	
6-RA-103-06	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-07	10.97	4.57	—	—	10.99	—	—	—	2006-03-08	
6-RA-103-08	10.97	4.57	—	—	10.97	—	—	—	2006-03-08	
6-RA-103-09	10.97	4.57	—	—	11.01	—	—	—	2006-03-08	
6-RA-103-10	10.97	4.57	—	—	11.02	—	—	—	2006-03-08	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-02	7.62	3.56	—	—	7.57	—	—	—	2001-03-08	
3-RV-101-03	7.62	3.56	—	—	7.52	—	—	—	2001-03-08	
3-RV-101-04	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	
3-RV-101-05	7.62	3.56	—	—	7.67	—	—	—	2001-03-08	
3-RV-101-06	7.62	3.56	—	—	7.59	—	—	—	2001-03-08	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	—	—	5.53	—	—	—	2001-03-08	
2-RV-102-02	5.54	2.79	—	—	5.59	—	—	—	2001-03-08	
2-RV-102-03	5.54	2.79	—	—	5.48	—	—	—	2001-03-08	
2-RV-102-04	5.54	2.79	—	—	5.58	—	—	—	2001-03-08	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	—	—	5.48	—	—	—	2006-03-08	
3-RF-101-02	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-03	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-04	5.49	2.41	—	—	5.46	—	—	—	2006-03-08	
3-RF-101-05	5.49	2.41	—	—	5.45	—	—	—	2006-03-08	
3-RF-101-06	5.49	2.41	—	—	5.52	—	—	—	2006-03-08	
3-RF-101-07	5.49	2.41	—	—	5.42	—	—	—	2006-03-08	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	—	—	3.92	—	—	—	2006-03-08	
2-RF-102-02	3.91	2.16	—	—	3.88	—	—	—	2006-03-08	
2-RF-102-03	3.91	2.16	—	—	3.93	—	—	—	2006-03-08	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2006-03-08	
2-RF-102-07	3.91	2.16	—	—	3.96	—	—	—	2006-03-08	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	—	—	6.07	—	—	—	2006-03-08	
4-LA-104-02	6.02	3.18	—	—	6.03	—	—	—	2006-03-08	
4-LA-104-03	6.02	3.18	—	—	5.96	—	—	—	2006-03-08	
4-LA-104-04	6.02	3.18	—	—	6.06	—	—	—	2006-03-08	
4-LA-104-05	6.02	3.18	—	—	6.04	—	—	—	2006-03-08	
4-LA-104-06	6.02	3.18	—	—	6.10	—	—	—	2006-03-08	
4-LA-104-07	6.02	3.18	—	—	6.01	—	—	—	2006-03-08	
4-LA-104-08	6.02	3.18	—	—	5.94	—	—	—	2006-03-08	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	—	—	8.51	—	—	—	2006-03-08	
4-RA-104-02	8.56	3.81	—	—	8.55	—	—	—	2006-03-08	
4-RA-104-03	8.56	3.81	—	—	8.54	—	—	—	2006-03-08	
4-RA-104-04	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-05	8.56	3.81	—	—	8.60	—	—	—	2006-03-08	
4-RA-104-06	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	
4-RA-104-07	8.56	3.81	—	—	8.59	—	—	—	2006-03-08	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	—	—	3.93	—	—	—	2006-03-08	
2-ST-101-02	3.91	2.29	—	—	3.90	—	—	—	2006-03-08	
2-ST-101-03	3.91	2.29	—	—	3.94	—	—	—	2006-03-08	
2-ST-101-04	3.91	2.29	—	—	3.82	—	—	—	2006-03-08	
2-ST-101-05	3.91	2.29	—	—	3.91	—	—	—	2006-03-08	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	—	—	5.42	—	—	—	2006-03-08	
3-CW-101-02	5.49	2.79	—	—	5.41	—	—	—	2006-03-08	
3-CW-101-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-101-04	5.49	2.79	—	—	5.46	—	—	—	2006-03-08	
3-CW-101-05	5.49	2.79	—	—	5.48	—	—	—	2006-03-08	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	—	—	5.50	—	—	—	2006-03-08	
3-CW-102-02	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	
3-CW-102-03	5.49	2.79	—	—	5.53	—	—	—	2006-03-08	
3-CW-102-04	5.49	2.79	—	—	5.51	—	—	—	2006-03-08	

Sign-off	Name	Function	Date	Signature
Prepared	Grace Okafor	UT technician, SNT-TC-1A UT Level II	1996-03-08	_____